

SpherIndent

An open-source Python software for calibration,
analysis and visualization of spherical
nanoindentation experiments

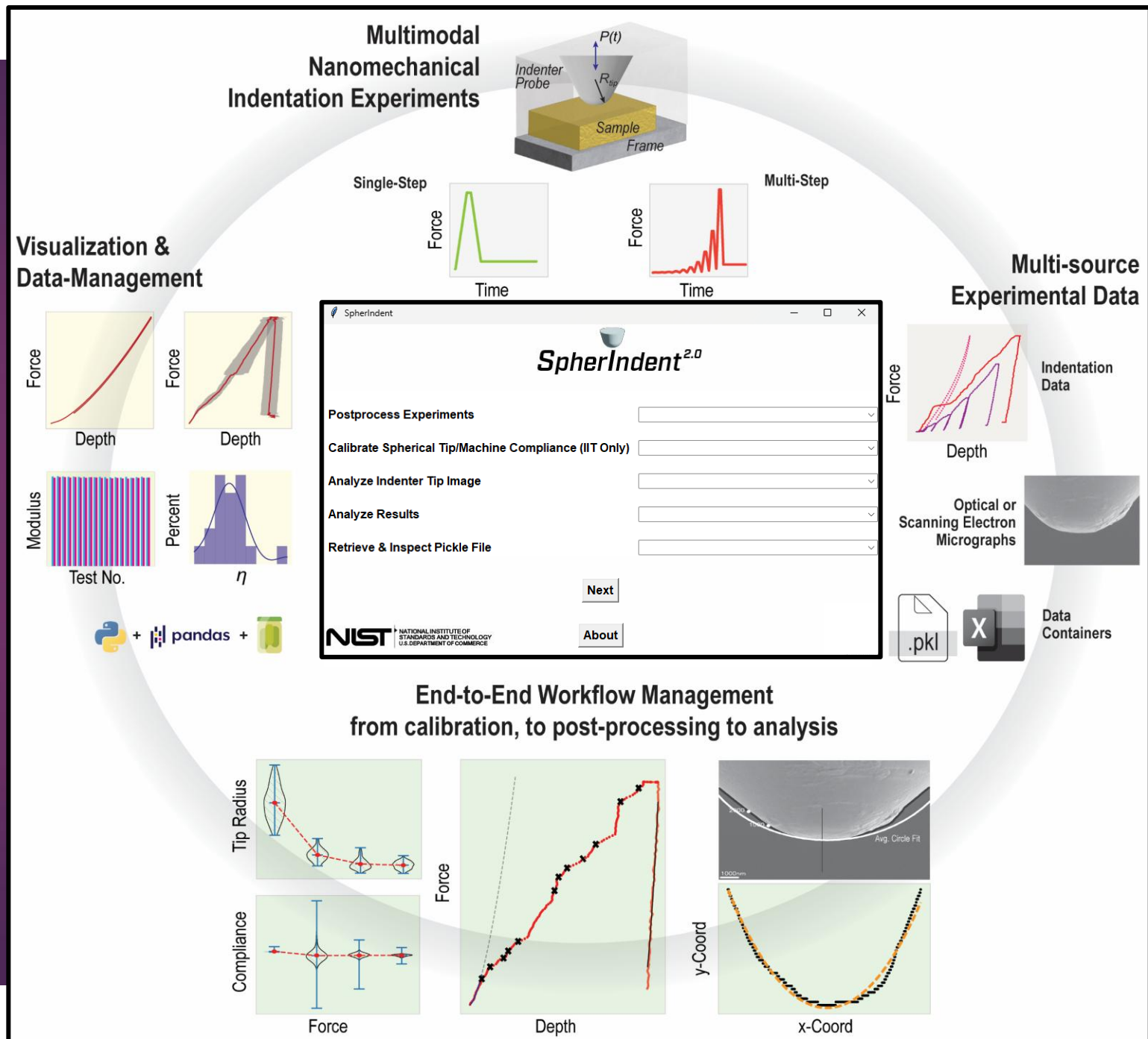
Quick Guide
Version: 2.0
(last updated Aug. 2026)

Useful Links

[Permanent Github Repository](#) | [Video Walkthrough](#) | [Peer-reviewed publication](#)

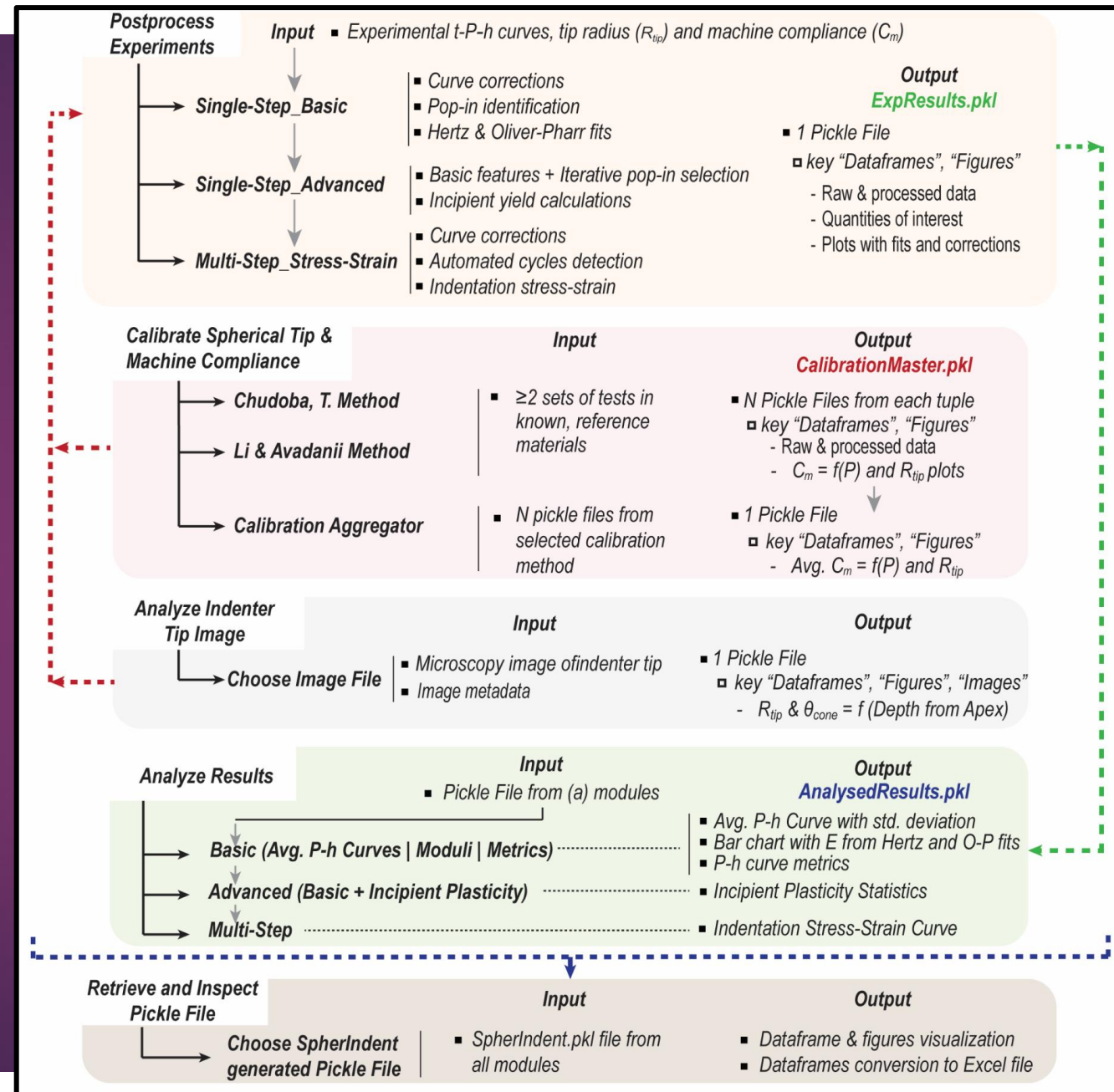
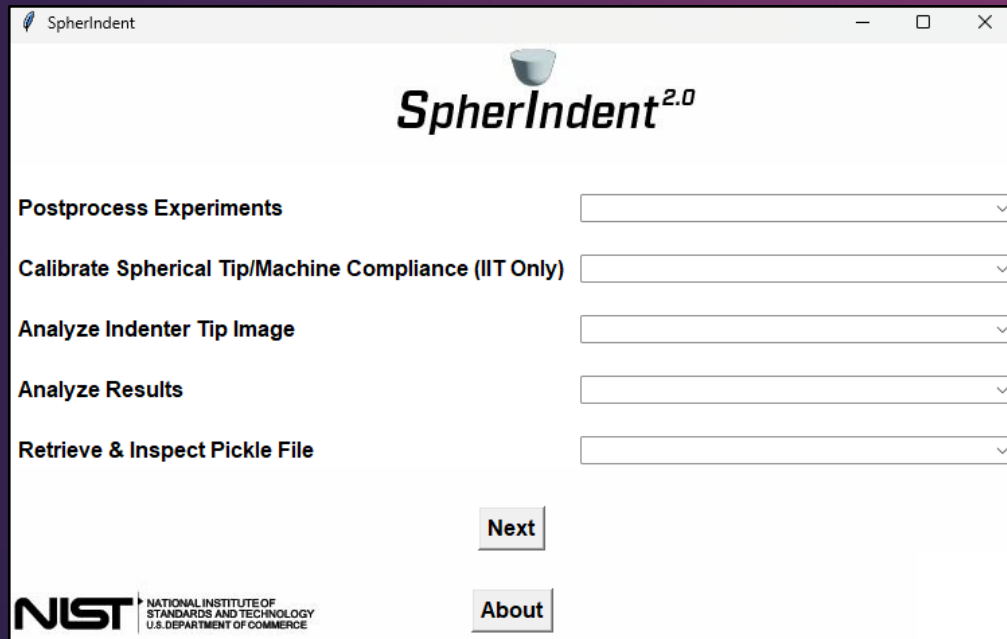
What's SpherIndent and Who is it Useful For?

- Modular, open-source package for analysis of quasi-static spherical indentation experiments conducted in **single-step** or **multi-step** loading sequences.
- Useful for:
 - Indentation practitioners
 - Researchers
 - Educators



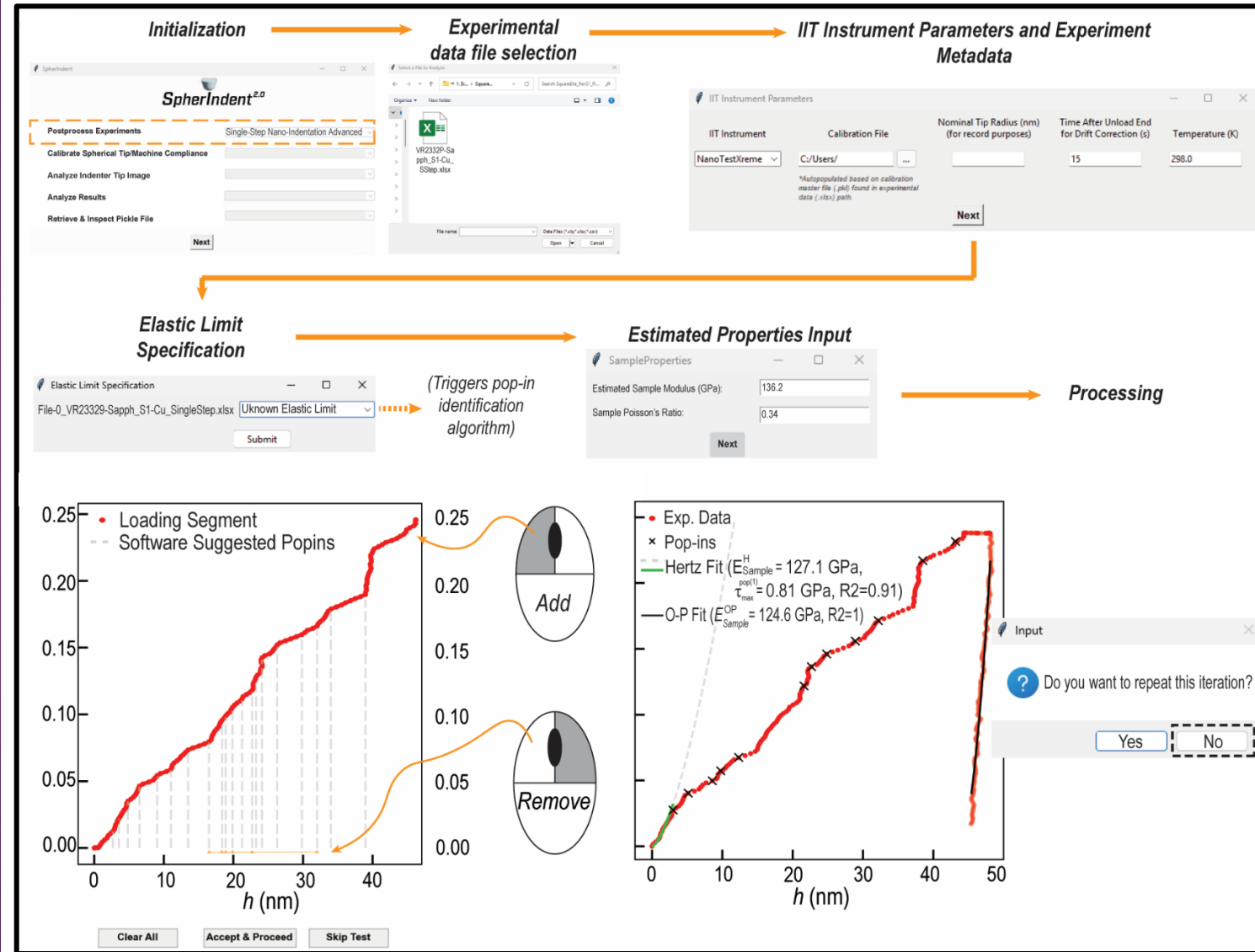
What can SpherIndent do?

- Five main modules with submodules that are accessible through dropdown menus, which can be run independently or sequentially.
- Data management via .pkl files.



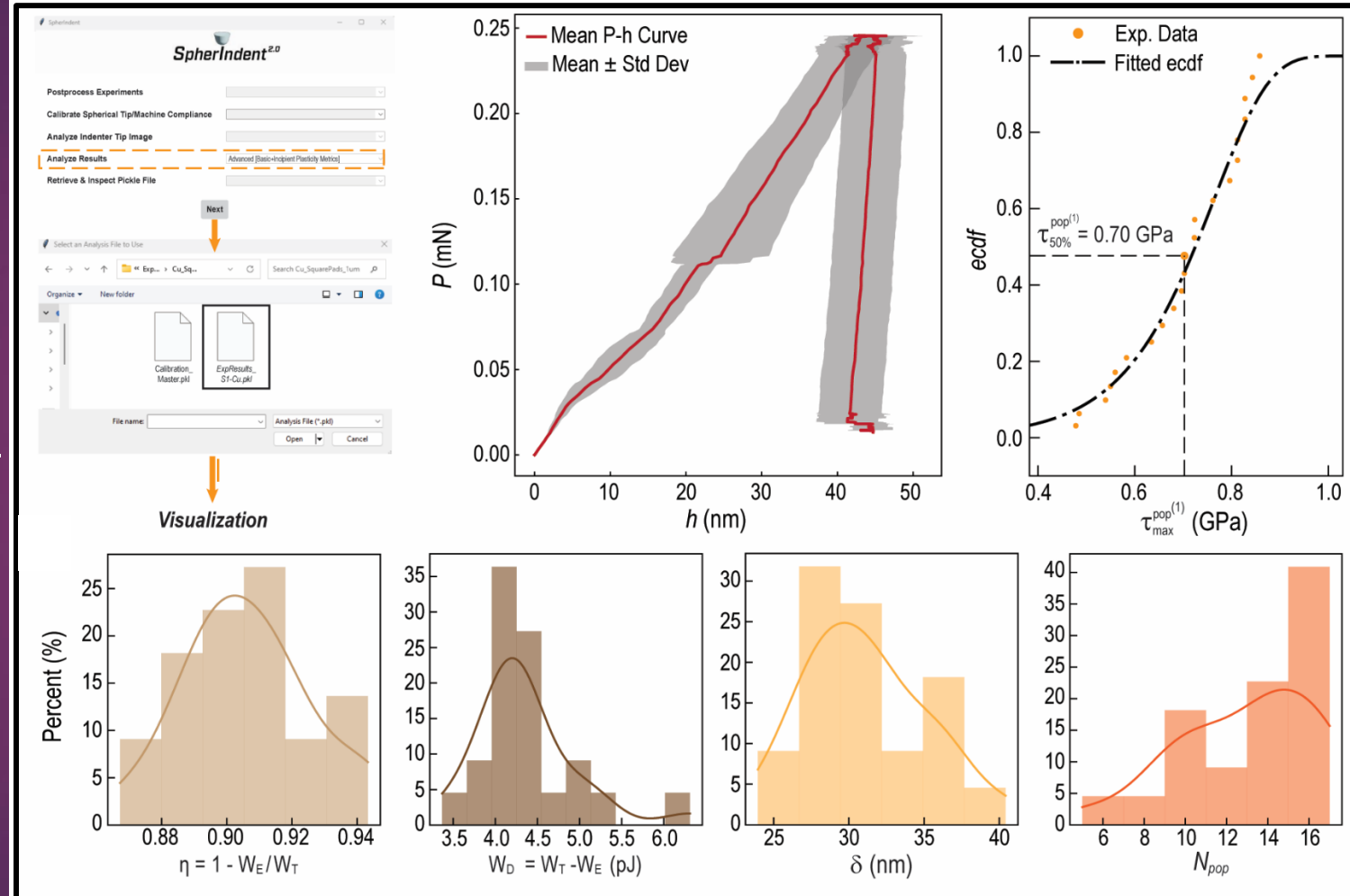
Single-Step Advanced

- User provides:
 1. An Excel file with n sheets containing Time-Displacement-Force experimental data.
 2. Experiment Metadata (equipment, thermal drift, calibration file, temperature, etc.)
- SpherIndent allows for:
 1. Interactive, software-aided pop-in (displacement burst) identification.
 2. Iterative 1st pop-in fitting



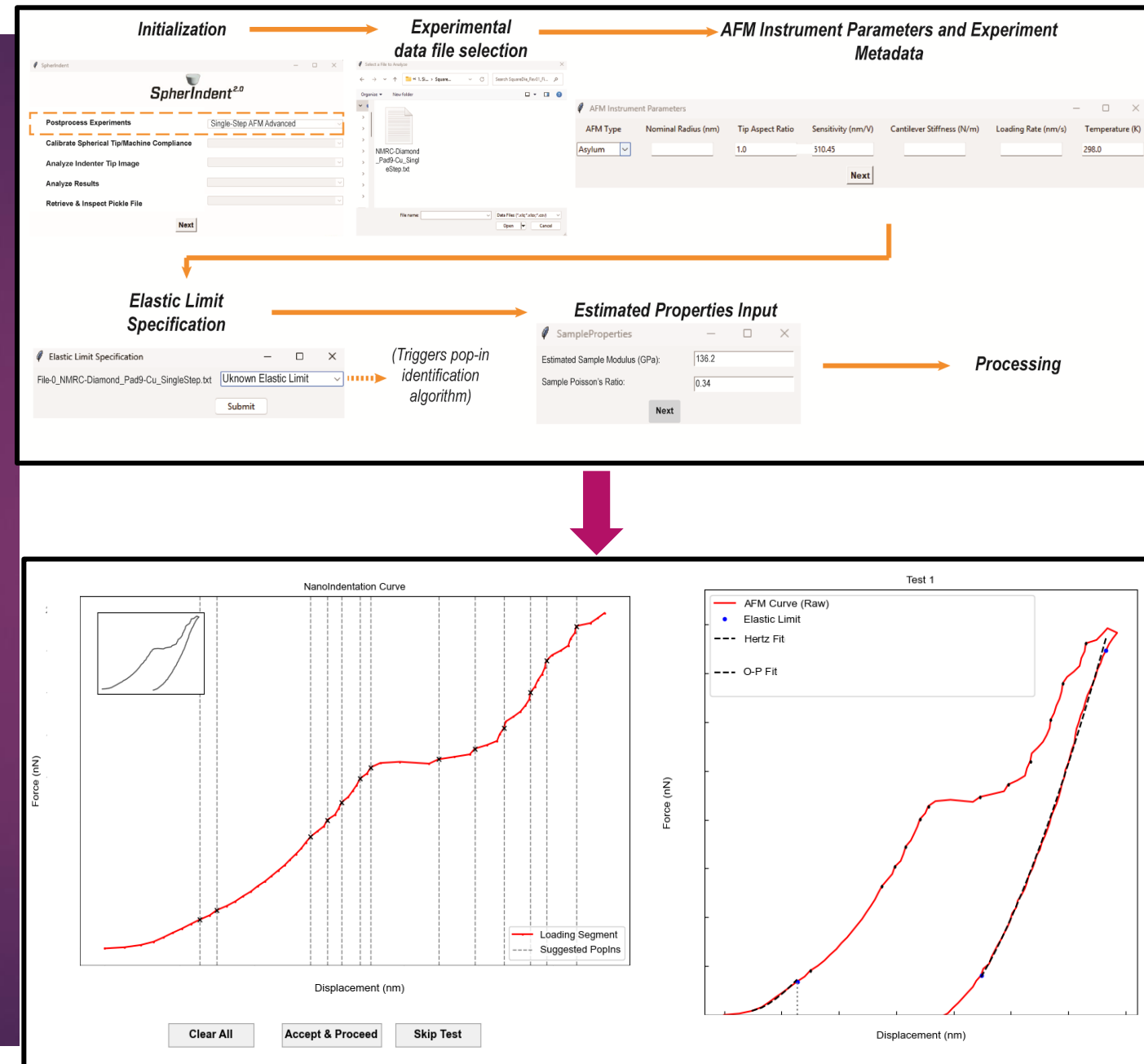
Single-Step Advanced

- SpherIndent outputs:
 - Hertzian and O-P fits in loading (upto 1st pop-in) and unloading segments, respectively.
 - Max. Shear stress at 1st pop-in.
 - Plasticity metrics.
- Further analysis can be conducted loading the .pkl file into the Analyze Results module.
- Statistical distributions



Single-Step Advanced AFM (Under development)

- User provides:
 1. A text file with n Voltage/Piezo-displacement experimental data.
 2. Experiment Metadata (equipment, sensitivity, cantilever stiffness, temperature, etc.)
- AFM inputs are text files from Force Maps containing deflection voltage for extension/retraction and piezo Z-sensor for extension/retraction

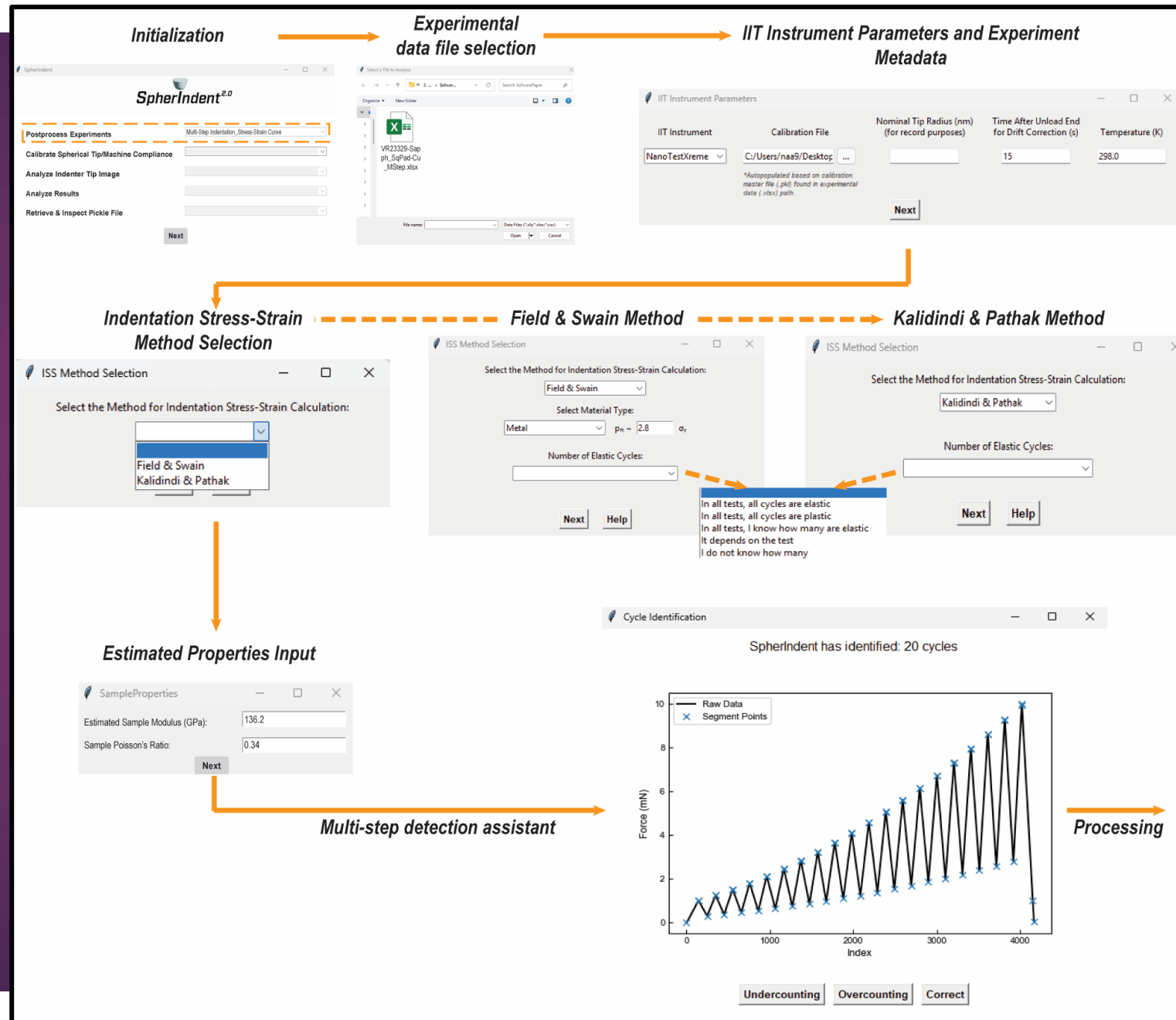


*Toy example

Multi-Step for Indentation Stress-Strain (ISS)

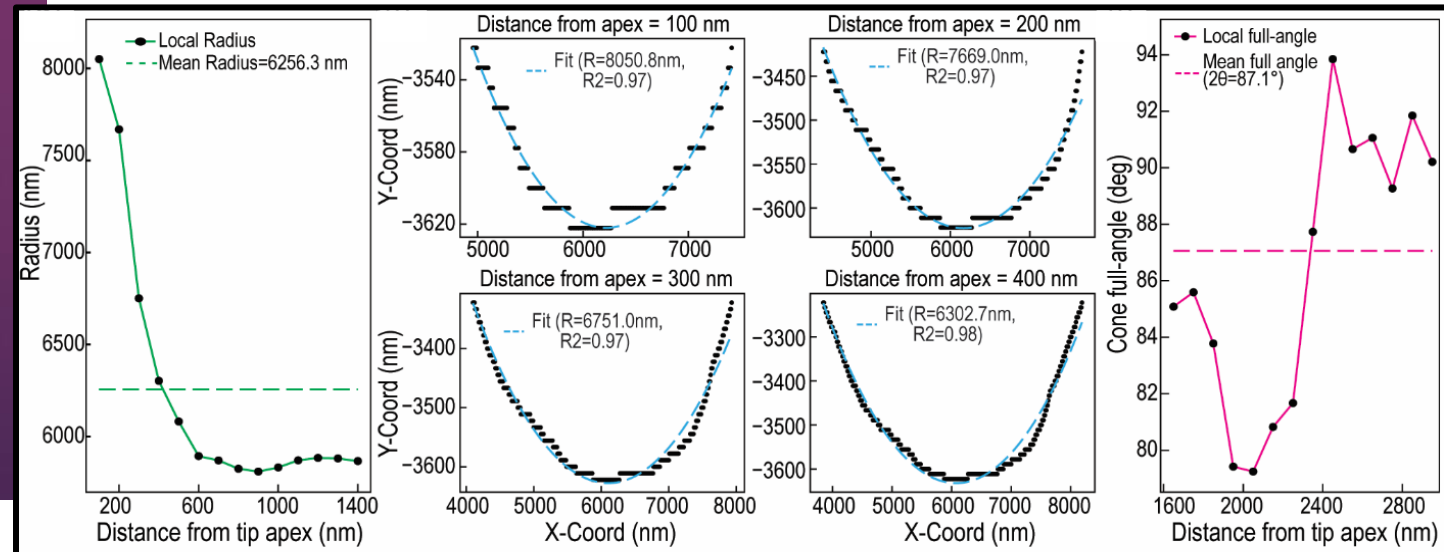
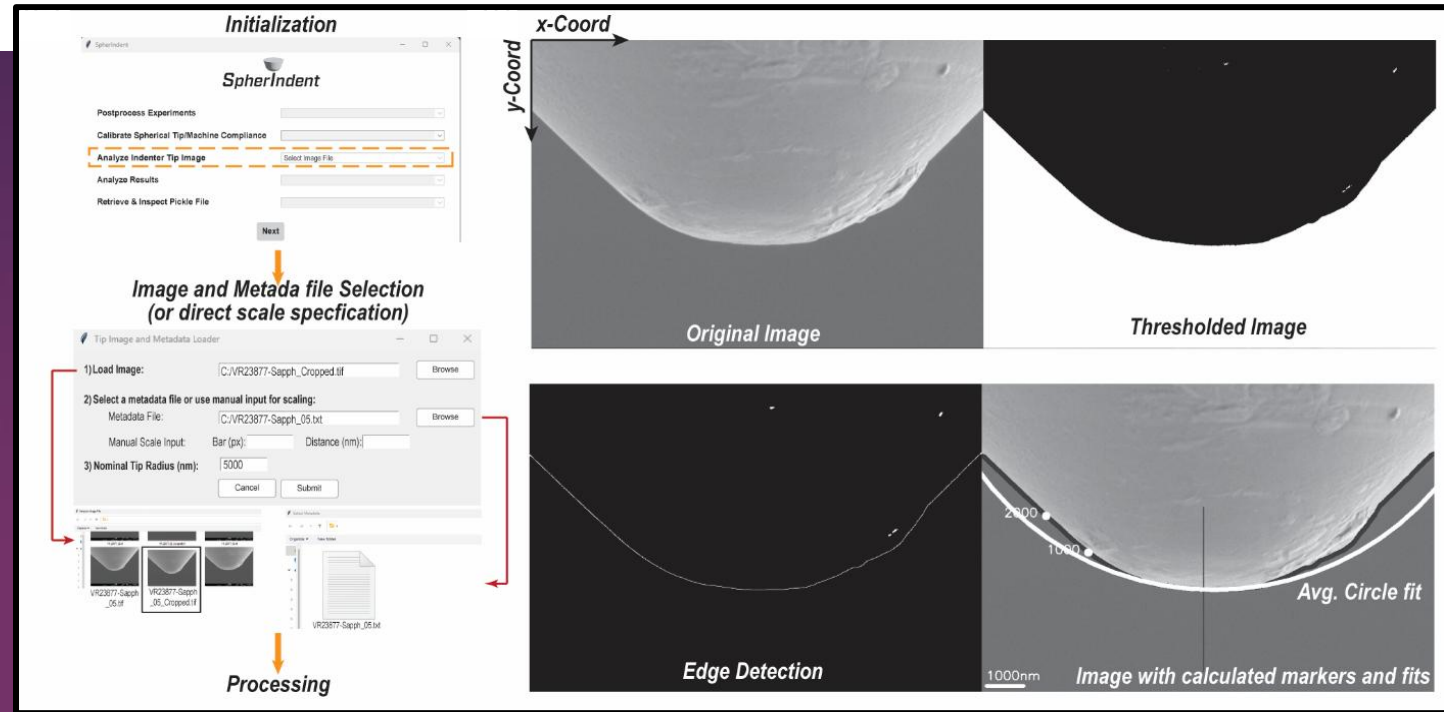
- User provides:

1. An Excel file with n sheets containing multi-step Time-Displacement-Force experimental data.
2. Experiment Metadata (equipment, thermal drift, calibration file, temperature, etc.).
3. ISS Method:
 1. Field & Swain (must provide material type).
 2. Kalidindi & Pathak.



Tip Image Analyzer

- User provides:
 1. A front view of the Nanoindenter tip.
 2. Image scale.
- SpherIndent outputs:
 1. Tip radius as a function of depth.
 2. Cone angle as a function of depth.
 3. Circle fits at prescribed depths.



Analyze Results

F-D Curve Averager

- User provides:
 - N loading curves and/or N unloading curves.
 - Simple copy-paste from a tabular source (e.g., MS Excel, Origin, etc.).
 - Option whether to average across displacements or forces.
- SpherIndent outputs:
 - Mean F-D curve with ± 1 standard deviation band.

Initialization

SpherIndent^{2.0}

Postprocess Experiments

Calibrate Spherical Tip/Machine Compliance

Analyze Indenter Tip Image

Analyze Results

Retrieve & Inspect Pickle File

Next

Simple paste from data in clipboard

Excel Data Picker

Paste Loading Curves (Disp-Force)

Proceed & Close

Visualize data and proceed

Excel Data Picker

Paste Loading Curves (Disp-Force)

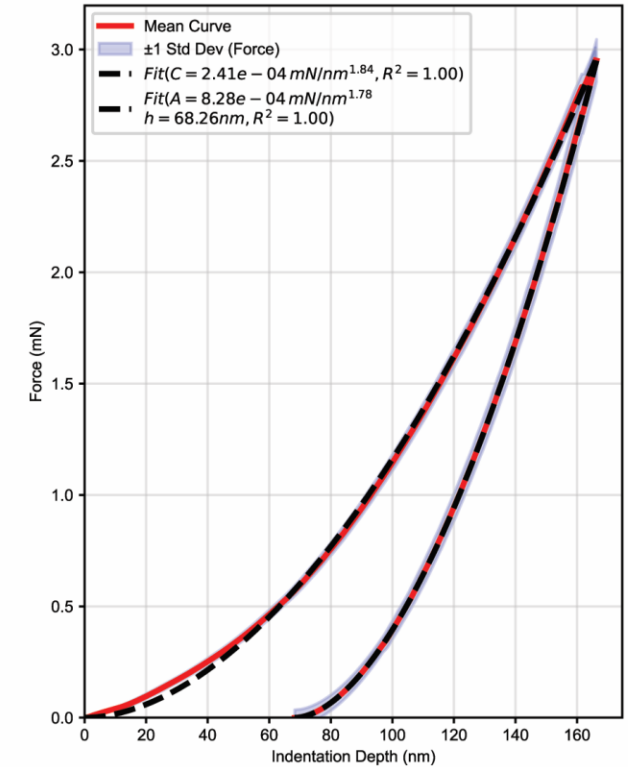
Proceed & Close

0	0	0	0	0	0	0	0	0	0
0.726127661	8.14059E-05	-0.126826601	8.10418E-05	-0.09531855	7.84335E-05	0.845196116	7.15711E-05	0.1830118	
0.720542755	8.19177E-05	-0.19508669	8.66322E-05	0.324136625	0.000114571	1.789181357	-6.15293E-06	1.0435529	
0.326235527	0.000184915	-0.201722915	0.000157788	0.049500627	0.000137013	2.200573776	3.06094E-05	1.9243791	
0.034308895	0.000279422	-0.189701002	0.000227355	-0.076410231	0.000217999	1.709677112	0.000141571	2.2904318	
2.0161763	0.011337229	1.330491732	0.015345561	0.140223511	0.000270854	2.211052033	0.011179918	1.9634629	
6.392519512	0.030807579	5.975300863	0.034793872	0.535492553	0.000308892	6.783967344	0.030634182	6.3361187	
11.20709016	0.050241869	10.24072767	0.054273324	3.550279534	0.016892121	11.89412503	0.049973569	11.700040	
14.40438202	0.069809158	13.68136852	0.07382064	8.888244531	0.036283431	15.42314084	0.069513636	15.595717	
17.63085721	0.089374047	16.54886922	0.09341499	13.27941942	0.055717267	18.68950733	0.089075265	18.949829	
20.56788348	0.108927441	19.45223422	0.11293599	16.47840289	0.075319754	21.89320162	0.108642041	21.731759	
23.18605521	0.128542418	22.56181265	0.132510555	19.30086596	0.094917947	24.60891348	0.128248985	24.870337	

Select Averaging Method

Averaging Method:

Average Forces



Output

Analyze Results

Residual Surface Averager

- User provides:
 - N residual surface profiles obtained by scanning probe microscopy (SPM).
 - Simple copy-paste from a tabular source (e.g., MS Excel, Origin, etc.).
- SpherIndent outputs:
 - Mean Surface Residual profile with ± 1 standard deviation band.

